
XY-Invariant Neural Networks for Sparse Timepix Particle Classification

Updated technical design for turning the repository into a runnable NN classification testbed

Primary design source:

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`xy_invariant_particle_report_project.zip`

Abstract

This report replaces the earlier report-first recommendation with an implementation-oriented design contract. Candidate particles should be represented as variable-size point sets

$$P = \{(x_i, y_i, t_i, e_i)\}_{i=1}^M,$$

where x, y are detector-plane coordinates, t is a time-like coordinate derived from ToA/FToA or frame metadata, and e is an energy-like feature such as $\log(1 + \text{ToT})$ until calibrated energy is available. The central correction is selective invariance: detector-plane rotation θ_{xy} must be computed and stored, but it must not define a class. A horizontal, vertical, or diagonal copy of the same track-like morphology should map to the same class embedding. Time pitch ϕ_t , shape, energy density, scale, and total energy remain classification-relevant.

The recommended baseline is a deterministic geometry front-end plus a compact canonical PointNet/DeepSets model called **XYInvariantParticleNet**. DBSCAN remains useful as an interpretable candidate generator, but clustering/classification after the encoder must operate only on z_{class} , never on a saved descriptor that includes θ_{xy} .

Main Design Contract

Same class: same morphology, similar energy-density profile, comparable scale, and similar time pitch, under any detector-plane rotation.

Stored metadata only: θ_{xy} and its reliability q_θ are saved for downstream physics, visualization, and direction statistics. They are excluded from class distance, HDBSCAN, DEC, VaDE, and classifier logits.

1 Data Inventory

The repository now has two data layers. The small tracked example is `data/matrix_dump_0001.txt`, a sparse JSON-style dump containing 198 matrices across 40 `sample` blocks. The large raw archive `raw_data.zip` is extracted locally under `local_data/raw/` and is intentionally gitignored.

The raw archive contains tab-separated `.t3pa` hit tables paired with `.t3pa.info` sidecars. Each raw row has columns `Index`, `Matrix Index`, `ToA`, `ToT`, `FToA`, and `Overflow`. The current parser decodes `Matrix Index` as row-major coordinates:

$$x = \text{index mod } 256, \quad y = \lfloor \text{index}/256 \rfloor.$$

The sidecars add acquisition metadata such as chipboard ID, high voltage, acquisition duration, threshold, Pixet version, and start time.

Table 1: Raw archive orientation generated by the repository indexer.

Item	Count / size	Notes
Tracked example dump	198 matrices	Sparse 256×256 calibrated-energy matrices
Raw files	711 <code>.t3pa</code> + 711 sidecars	Six run folders in <code>local_data/raw/</code>
Uncompressed raw bytes	1.426 GB	Large data remains outside git
Raw hit rows	44,725,206	Counted directly from <code>.t3pa</code> rows
Index artifacts	CSV + Markdown	<code>data/raw_data_index.csv</code> , <code>docs/raw-data-index.md</code>

2 Target Product

The measured input is not yet a particle class. It is a set of detector hits. The useful reconstruction chain should keep these layers separate:

1. hit filtering or denoising;
2. clustering or instance segmentation into particle candidates;
3. candidate-level morphology/energy classification;
4. optional association across time, views, or chipboards;
5. run-level summaries and physics validation.

The first practical deliverable is therefore:

raw hits $\rightarrow$ candidate clustering $\rightarrow$ XY-invariant candidate descriptor and class confidence.

The returned candidate record should include at least source path or `sample/set_index`, candidate ID, hit list or mask, total energy/ToT, size, eccentricity or PCA ratios, z_{class} , class confidence or cluster ID, θ_{xy} , q_θ , and ϕ_t .

3 Selective XY Invariance

Detector-plane azimuth is a nuisance factor for morphology classification. A straight object at 0° , 45° , and 90° should not become three classes. Conversely, the time axis is not another spatial axis. The pitch between the spatial trajectory and time can encode a physically meaningful difference and should remain available to the class descriptor.

Table 2: Which factors may affect classification or clustering.

Factor	Use in class distance?	Reason
Shape latent z_{shape}	Yes	Defines point-like, blob-like, track-like, curved, shower-like, or fragmented morphology
Energy-density latent $z_{density}$	Yes	Separates shapes with different ToT/energy profiles
Scale z_{scale}	Yes, configurable	Length, width, duration, and hit count may be class-relevant
Time pitch z_{tilt} / ϕ_t	Yes	Time is physically special and should not be mixed with detector-plane rotation
Energy summary z_{energy}	Yes	Needed for energetic taxonomy and downstream validation
Detector-plane azimuth θ_{xy}	No	Store and analyze, but never use to split morphology classes
Angle reliability q_θ	No	Metadata indicating whether θ_{xy} is meaningful

4 Geometry Front-End

For a candidate $P = \{(x_i, y_i, t_i, e_i)\}_{i=1}^M$, compute an energy-weighted centroid and covariance. Let $w_i = e_i + \epsilon$ and use a time scale λ_t to place the time feature in a controlled numerical range:

$$r_i = (x_i - \bar{x}, y_i - \bar{y}, \lambda_t(t_i - t_{min})).$$

The weighted covariance eigenvectors provide two useful angles:

$$\theta_{xy} = \text{atan2}(v_y, v_x), \quad \phi_t = \text{atan2}\left(v_t, \sqrt{v_x^2 + v_y^2}\right).$$

For compact or one-hit candidates, detector-plane angle can be meaningless. Store a reliability score:

$$q_\theta = \frac{\lambda_{xy,1} - \lambda_{xy,2}}{\lambda_{xy,1} + \lambda_{xy,2} + \epsilon}.$$

For the class branch, rotate only x, y into a canonical frame:

$$\begin{pmatrix} x'_i \\ y'_i \end{pmatrix} = R(-\theta_{xy}) \begin{pmatrix} x_i - \bar{x} \\ y_i - \bar{y} \end{pmatrix}.$$

The original θ_{xy} is preserved in metadata, not in the clustering vector.

5 XYInvariantParticleNet

The first neural model should be deliberately modest: a PointNet/DeepSets encoder over canonicalized variable-size point sets [Qi et al., 2017, Zaheer et al., 2017]. A stronger local model can later add EdgeConv or point-transformer blocks [Wang et al., 2019, Zhao et al., 2021, Robles et al., 2025].

Table 3: Reference model contract.

Component	Role
Input candidate	Variable-size set $[M, F]$ with canonicalized x', y' , relative time, and $\log(1 + \text{ToT})$ or calibrated energy
Geometry front-end	Computes centroid, scale, weighted PCA, θ_{xy} , q_θ , ϕ_t , and energy-density profiles
Hit MLP	Shared per-hit encoder for any number of hits
Pooling	Max, mean, and attention pooling produce a fixed-length candidate vector
Class latent	$z_{class} = [z_{shape}, z_{density}, z_{scale}, z_{tilt}, z_{energy}]$; excludes θ_{xy}
Pose metadata head	Predicts or validates θ_{xy} and q_θ for saved metadata only
Clustering layer	HDBSCAN, DEC, IDEC, VaDE, GMM, or prototype clustering over z_{class} only [McInnes et al., 2017, Xie et al., 2016, Guo et al., 2017, Jiang et al., 2017]

The implementation rule is:

$$D(P_a, P_b) = D(z_{class}^a, z_{class}^b) \quad \text{and never} \quad D(z_{saved}^a, z_{saved}^b).$$

If θ_{xy} leaks into D , a later clustering algorithm will split horizontal, vertical, and diagonal copies of the same physical morphology.

6 Training Without Labels

The repository should first support self-supervised or weakly supervised learning. The useful loss family is:

$$\mathcal{L} = \mathcal{L}_{profile} + \alpha \mathcal{L}_{xy-inv} + \beta \mathcal{L}_{xy-pose} + \gamma \mathcal{L}_{tilt} + \delta \mathcal{L}_{density}.$$

The core invariance loss rotates the detector plane and requires the class embedding to stay stable:

$$\mathcal{L}_{xy-inv} = \|z_{class}(P) - z_{class}(R_{\Delta\theta}P)\|_2^2.$$

The pose head can still be trained to shift by the known augmentation angle:

$$\hat{\theta}(R_{\Delta\theta}P) \approx \hat{\theta}(P) + \Delta\theta.$$

For undirected line-like candidates, use double-angle encoding $(\cos 2\theta, \sin 2\theta)$ so θ and $\theta + \pi$ represent the same axis.

Useful self-supervised targets include longitudinal/radial energy profiles, ToT quantiles, PCA eigenvalue ratios, hit count, total ToT, and time span. A masked point autoencoder or profile autoencoder can later be added, following the same sparse point-set logic [Young et al., 2025].

7 Baselines and Roadmap

Stage 0: raw-data orientation. Extract the raw archive into `local_data/raw/`, generate `docs/raw-data-index.md`, and verify row counts, metadata, and run folders.

Stage 1: deterministic XY-invariant descriptor. Compute weighted PCA, energy profiles, scale, energy summaries, and ϕ_t . Cluster these descriptors with HDBSCAN or DBSCAN, excluding θ_{xy} . This is the sanity check against which the NN must improve.

Stage 2: canonical PointNet/DeepSets. Train XYInvariantParticleNet on DBSCAN candidates or curated candidate windows using rotation-invariance, pose, and profile losses. Report rotation consistency and descriptor leakage.

Stage 3: stronger point models. Add EdgeConv, GravNet-like latent neighborhoods, or point transformers when local shape, branches, and overlapping deposits require more capacity [Qasim et al., 2019, Qu and Gouskos, 2020, Qu et al., 2022].

Stage 4: learned clustering. If simulation or hit-level truth becomes available, replace the DBSCAN candidate generator with object condensation or a related instance-segmentation model [Kieseler, 2020, Qasim et al., 2021, Garcia et al., 2026].

8 Validation

Accuracy alone is not sufficient. The required tests are:

- **rotation consistency:** variance of $z_{class}(R_\alpha P)$ over random α ;
- **pose accuracy:** circular error of predicted or stored θ_{xy} ;
- **cluster invariance:** fraction of rotated copies assigned to the same cluster;
- **descriptor leakage:** correlation or mutual information between cluster ID and θ_{xy} should be low;
- **tilt sensitivity:** ϕ_t changes can separate groups when intended;
- **interpretability:** cluster medoids should differ by shape, energy density, scale, or time pitch, not by detector-plane azimuth.

9 Conclusion

The repository should treat raw Timepix rows and sparse matrix dumps as measured detector hits, not ordinary images. The immediate implementation target is a runnable baseline that can parse raw data, build candidate descriptors, run DBSCAN as a bootstrap, and train a compact point-set encoder. The governing design principle is selective invariance: compute and preserve detector-plane orientation, but do not let it define the morphology class. This makes the system debuggable today and keeps a clean path toward stronger graph, transformer, or object-condensation models when labels or simulation truth become available.

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